

Hamilton Pinching Conjecture

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1 Background

We first state the theorem.

Theorem 1.1 (Hamilton pinching conjecture). *Suppose (M, g_0) is a complete Riemannian manifold satisfying*

$$R_{g_0} - \varepsilon \operatorname{Scal}_{g_0} I \in C_{\text{PIC1}}, \quad \varepsilon \in \left(0, \frac{1}{n(n-1)}\right).$$

Then (M, g_0) is either compact or flat.

The following partial results were known previously.

1. Brendle and Schoen proved the conjecture under the additional assumptions that the curvature is PIC2-pinched, $\operatorname{Scal} > 0$, and the sectional curvature is uniformly bounded.
2. Lee and Topping proved the three-dimensional case, as well as the higher-dimensional case under the assumption $\operatorname{Rm} \in C_{\text{PIC2}}$.
3. Deruelle, Schulze, and Simon proved the case in which $\operatorname{AVR} > 0$.

1.1 Notation

We define the cone

$$\begin{aligned} \check{C}(s) := \Big\{ R \in C_B(\mathbb{R}^n) : R_{1313} + \lambda^2 R_{1414} + \mu^2 R_{2323} + \lambda^2 \mu^2 R_{2424} - 2\lambda\mu R_{1234} \\ + \frac{1}{s}(1 - \lambda^2)(1 - \mu^2) \operatorname{Scal}(R) \geq 0, \quad \lambda, \mu \in [0, 1] \Big\}. \end{aligned}$$

For

$$a = b + \frac{1}{2}(n-2)b^2,$$

set

$$\tilde{C}(b, s) := \ell_{a,b}(\check{C}(s)).$$

We can verify that both $\tilde{C}(b, s)$ and $\hat{C}(s)$ have the following properties:

1. they are closed, convex, and $O(n)$ -invariant;
2. they are transversely invariant under Hamilton's ODE;
3. every nonzero algebraic curvature tensor in either cone has positive scalar curvature;
4. I lies in the interior of the cone.

These statements hold provided

$$0 < b < \Upsilon_n := \frac{\sqrt{2n(n-2)+4}-2}{n(n-2)}.$$

These cones are useful because they allow us to translate the pinching condition into a cone condition.

Lemma 1.2 (Pinching-to-cone conversion). *Suppose $R \in C_B(\mathbb{R}^n)$ is PIC1-pinched. Then there exist $s_0 = s_0(n, \varepsilon_0)$ and $b = b(n, \varepsilon_0)$ such that*

$$R \in \tilde{C}(b, s_0).$$

2 Outline of the Proof

2.1 An existence theorem for Ricci flow

Theorem 2.1 (Long-time existence). *For any $n \geq 4$ and $\varepsilon_0 \in \left(0, \frac{1}{n(n-1)}\right)$, there exist $a_0 > 0$ and $\varepsilon'_0 \in \left(0, \frac{1}{n(n-1)}\right)$ with the following property. Suppose (M, g_0) is a complete noncompact manifold satisfying*

$$R_{g_0} - \varepsilon_0 \operatorname{Scal}(R_{g_0})I \in C_{\text{PIC1}}$$

on M . Then there exists a smooth complete Ricci flow solution $g(t)$ on $M \times [0, +\infty)$ such that, for every $t > 0$,

1. $R_{g(t)} - \varepsilon'_0 \operatorname{Scal}(R_{g(t)})I \in C_{\text{PIC1}};$
2. $|R_{g(t)}| \leq a_0 t^{-1}.$

This theorem is due to Lee and Topping.

2.2 The pinching conjecture under positive asymptotic volume ratio

Deruelle, Schulze, and Simon proved the Hamilton pinching conjecture under the assumption of maximal volume growth using the theory of Cheeger-Colding.

2.3 A new lifting argument

The new argument in the recent paper by the five authors combines the two results above to prove the conjecture.

In our talk, we introduce the main ideas behind the first and third parts.

3 The Existence Theorem

We begin by restating [theorem 2.1](#).

Theorem 2.1 (restated). *For any $n \geq 4$ and $\varepsilon_0 \in \left(0, \frac{1}{n(n-1)}\right)$, there exist $a_0 > 0$ and $\varepsilon'_0 \in \left(0, \frac{1}{n(n-1)}\right)$ with the following property. Suppose (M, g_0) is a complete noncompact manifold satisfying*

$$R_{g_0} - \varepsilon_0 \operatorname{Scal}(R_{g_0})I \in C_{\text{PIC1}}$$

on M . Then there exists a smooth complete Ricci flow solution $g(t)$ on $M \times [0, +\infty)$ such that, for every $t > 0$,

1. $R_{g(t)} - \varepsilon'_0 \operatorname{Scal}(R_{g(t)})I \in C_{\text{PIC1}};$
2. $|R_{g(t)}| \leq a_0 t^{-1}.$

We first take the existence assertion for granted and derive a curvature estimate.

Proposition 3.1 (Pointwise curvature estimate). *Suppose M^n is a noncompact connected manifold with $n \geq 4$, and $g(t)$, $t \in [0, T]$, is a smooth solution to the Ricci flow on M . Suppose that for some $x_0 \in M$, $s_0 > 0$, and $b \in (0, \Upsilon_n)$, we have*

- (i) $B_{g(t)}(x_0, 1) \Subset M$ for all $t \in [0, T];$
- (ii) $R_{g(t)} + I \in \tilde{C}(b, s_0)$ on $B_{g(t)}(x_0, 1)$ for all $t \in [0, T].$

Then there exist $C_0 = C_0(n, b, s_0) > 0$ and $S_1 = S_1(n, b, s_0) > 0$ such that

$$|R_{g(t)}(x_0)| \leq C_0 t^{-1}$$

for every $t \in (0, S_1 \wedge T]$.

Sketch of proof

We argue by contradiction. Suppose the conclusion is false. Then we can find a sequence of manifolds $(M_k, g_k(t))$, with $t \in [0, t_k]$, and numbers $a_k \rightarrow \infty$ such that

1. $B_{g_k(t)}(x_k, 1) \Subset M_k$ for $t \in [0, t_k]$;
2. $R_{g_k(t)} + I \in \tilde{C}(b, s_0)$ on $B_{g_k(t)}(x_k, 1)$ for $t \in [0, t_k]$;
3. $|R_{g_k(t)}(x_k)| < a_k t^{-1}$ for $t \in (0, t_k]$;
4. $|R_{g_k(t_k)}(x_k)| = a_k t_k^{-1}$.

By choosing t_k small, we may also assume that $a_k t_k \rightarrow 0$.

This estimate is centered at x_k , but we need a local estimate. A standard point-picking argument gives, for all sufficiently large k , a dimensional constant $\beta(n) > 0$, times $\tilde{t}_k \in (0, t_k]$, and points

$$\tilde{x}_k \in B_{g_k(\tilde{t}_k)}\left(x_k, \frac{3}{4} - \frac{1}{2}\beta\sqrt{a_k\tilde{t}_k}\right)$$

such that, writing

$$Q_k := |R_{g_k(\tilde{t}_k)}(\tilde{x}_k)|,$$

we have

$$|R_{g_k(t)}(x)| \leq 4|R_{g_k(\tilde{t}_k)}(\tilde{x}_k)| = 4Q_k \tag{3.1}$$

whenever

$$d_{g_k(\tilde{t}_k)}(x, \tilde{x}_k) < \frac{1}{8}\beta a_k Q_k^{-1/2}, \quad \tilde{t}_k - \frac{1}{8}a_k Q_k^{-1} \leq t \leq \tilde{t}_k,$$

where $\tilde{t}_k Q_k \geq a_k \rightarrow +\infty$.

Now consider the parabolic rescaling centered at $(\tilde{x}_k, \tilde{t}_k)$,

$$\tilde{g}_k(t) := Q_k g_k(\tilde{t}_k + Q_k^{-1}t), \quad t \in \left[-\frac{1}{8}a_k, 0\right].$$

It satisfies

- (a) $|R_{\tilde{g}_k(0)}(\tilde{x}_k)| = 1$;
- (b) $|R_{\tilde{g}_k(t)}| \leq 4$ on $B_{\tilde{g}_k(0)}(\tilde{x}_k, \frac{1}{8}\beta a_k) \times [-\frac{1}{8}a_k, 0]$;
- (c) $R_{\tilde{g}_k(t)} + Q_k^{-1}I \in \tilde{C}(b, s_0)$ on the same parabolic neighborhood.

Always keep in mind that $a_k, Q_k \rightarrow +\infty$. If we can take a limit of these manifolds, then we obtain an ancient flow.

Claim 3.2 (Cone improvement). *For any $L, \varepsilon > 0$ and $s \geq s_0$, there exists $N \in \mathbb{N}$ such that, for all $k > N$,*

$$R_{\tilde{g}_k(t)} + \varepsilon I \in \tilde{C}(b, s)$$

on

$$B_{\tilde{g}_k(0)}(\tilde{x}_k, L) \times [-L^2, 0].$$

Proof. Let $\mathcal{S}$ be the set of $s' \in [s_0, +\infty)$ such that, for all $L > 0$ and $\varepsilon \in (0, 1)$, we can find $N \in \mathbb{N}$ for which, whenever $k > N$,

$$R_{\tilde{g}_k(t)} + \varepsilon I \in \tilde{C}(b, s')$$

on

$$B_{\tilde{g}_k(0)}(\tilde{x}_k, L) \times [-L^2, 0].$$

Clearly $s_0 \in \mathcal{S}$ by property (c) above. We show that $\mathcal{S}$ is both open and closed, and hence that $\mathcal{S} = [s_0, +\infty)$. $\square$

The claim shows that the limiting process sends the curvature into $\ell_{a,b}(C_{\text{PIC2}}) \subset C_{\text{PIC2}}$.

Since the family of cones $\hat{C}(s)$ connects C_{PIC2} with $\mathbb{R}_+I$, we can choose s sufficiently large and s_0 sufficiently small so that

$$\tilde{C}(b, s) \subset \hat{C}(s_0).$$

By Claim 3.2, and because the metric lies in some $\hat{C}(s_0)$ for all sufficiently large k , we can repeat the argument in the claim to conclude that the limit actually lies in $\mathbb{R}_+I$. Thus the limit is locally a space form.

There is still an obstacle to taking this limit: the Cheeger–Gromov–Hamilton compactness theorem requires a lower bound for the injectivity radius. This is not a serious problem, because we can lift the ball to the tangent space by means of the exponential map.

To get the global existence, we first need to obtain the following local result.

Theorem 3.3 (Local existence with scale-invariant estimates). *For any $s_0 > 0$, $n \geq 4$, and $b \in (0, \Upsilon_n)$, there exist $a_0 = a_0(n, b, s_0) > 0$ and $T = T(n, b, s_0) > 0$ with the following property. Suppose (M^n, g_0) is a manifold such that*

- (a) $B_{g_0}(p, 4) \Subset M$;
- (b) $R_{g_0} \in \tilde{C}(b, s_0)$ on $B_{g_0}(p, 4)$.

Then there exists a smooth Ricci flow solution $g(t)$ on $B_{g_0}(p, 1) \times [0, T]$ such that $g(0) = g_0$ and

- (i) $R_{g(t)} + I \in \tilde{C}(b, s_0)$;
- (ii) $|R_{g(t)}| \leq a_0 t^{-1}$.

The key difficulty is that the parameter T must not depend on the curvature of M .

There is, in fact, only a small gap between this local result and the global result. We use rescaling. For the initial data (M, g_0) , define

$$h_{i,0} = R_i^{-2} g_0.$$

Applying theorem 3.3, we obtain a local Ricci flow $h_i(t)$ on $B_{h_{i,0}}(p, 1) \times [0, T]$. The parabolically rescaled flow

$$g_i(t) := R_i^2 h_i(R_i^{-2} t)$$

is defined on

$$B_{g_0}(p, R_i) \times [0, T R_i^2]$$

and satisfies

$$R_{g_i(t)} + R_i^{-2} I \in \tilde{C}(b, s_0).$$

Taking a limit gives the immortal Ricci flow $g(t)$ and the desired pinching condition

$$R_{g(t)} \in \tilde{C}(b, s_0).$$

We now prove the local result by a bootstrapping argument. The first ingredient is a result similar to what we want, except that the existence time depends on the initial curvature and there is a multiplier $\Lambda > 1$.

Proposition 3.4 (Short-time local existence). *For $n \geq 2$, there exist constants $\alpha \in (0, 1]$ and $\Lambda > 1$, depending only on n , with the following property. Suppose (N^n, h_0) is a smooth manifold, not necessarily complete, satisfying*

$$|R_{h_0}| \leq \rho^{-2}$$

throughout N , for some $\rho > 0$. Then there exists a smooth Ricci flow $h(t)$ on N for $t \in [0, \alpha\rho^2]$ such that

- (i) $h(0) = h_0$ on $N_\rho := \{x \in N : B_{h_0}(x, \rho) \Subset N\}$;
- (ii) $|R_{h(t)}| \leq \Lambda\rho^{-2}$ throughout $N_\rho \times [0, \alpha\rho^2]$.

This proposition is not hard to prove. To be specific, we can modify the metric locally and let the metric complete. Then due to the result of W.X.Shi, we get the local short-time existence.

This proposition helps us to start the bootstrapping. What's more, we also need the following maximum principle.

Lemma 3.5 (Local persistence of the cone condition). *Suppose $n \geq 4$ and $(M^n, g(t))$, $t \in [0, T]$, is a smooth solution to the Ricci flow with $g(0) = g_0$. Suppose that for some $x_0 \in M$,*

- (i) $B_{g_0}(x_0, 1) \Subset M$;
- (ii) $R_{g_0} \in \tilde{C}(b, s_0)$ on $B_{g_0}(x_0, 1)$ for some $s_0 > 0$ and $b \in (0, \Upsilon_n)$;
- (iii) $|R_{g(t)}| \leq c_0 t^{-1}$ on $B_{g_0}(x_0, 1) \times (0, T]$ for some $c_0 < \infty$.

Then there exists $S_0 = S_0(n, c_0, b, s_0) > 0$ such that, for every $t \in [0, T \wedge S_0]$,

$$R_{g(t)}(x_0) + I \in \tilde{C}(b, s_0).$$

The bootstrap process can now be summarized as follows.

- I. Use Proposition 3.4 to initiate a local flow for a short time interval $[0, \alpha\rho^2]$.
- II. Use the curvature estimate to obtain $|\text{Rm}| \leq a_0/t$.
- III. Use Lemma 3.5 to obtain $R_{g(t)} + I \in \tilde{C}(b, s_0)$ for a short time.
- IV. Use Proposition 3.1 to improve the estimate to $|\text{Rm}| < c_0/t$.
- V. Apply Proposition 3.4 again on the smaller region to continue the flow.

Repeating these five steps lengthens the time interval while narrowing the spatial region. In this way, the eventual existence time can be made independent of the initial curvature.

4 The New Argument

The new argument is carried out after pulling back by the exponential map. We need two technical lemmas based on distance comparison along the Ricci flow.

Lemma 4.1 (Shrinking balls lemma). *Suppose $(M, g(t))$ is a Ricci flow for $t \in [0, T]$ on a manifold M of dimension n . There exists $\beta = \beta(n) \geq 1$ with the following property. Suppose $x_0 \in M$, $B_{g(0)}(x_0, r) \Subset M$ for some $r > 0$, and, for some $c_0 > 0$,*

$$\text{Ric}_{g(t)} \leq \frac{(n-1)c_0}{t}$$

on $B_{g(0)}(x_0, r)$ for every $t \in (0, T]$. Then

$$B_{g(0)}(x_0, r) \supset B_{g(t)}(x_0, r - \beta\sqrt{c_0 t})$$

for every $t \in [0, T]$.

Lemma 4.2 (Eternally shrinking balls lemma). *Suppose $(M, g(t))$ is a Ricci flow, not necessarily complete, for $t \in [0, T]$ on a manifold M of dimension n . Suppose $x_0 \in M$, $R > 0$, and $c_0 > 0$ satisfy*

1. $B_{g(0)}(x_0, R) \Subset M$;
2. $\text{Ric}_{g(t)} \leq (n-1)c_0/t$ on $B_{g(0)}(x_0, R)$ for every $t \in (0, T]$.

Then there exists $r > 0$, depending only on n, c_0, R , and T (or merely on an upper bound for T), such that, for every $t \in [0, T]$,

$$B_{g(0)}(x_0, R) \supset B_{g(t)}(x_0, r).$$

This two lemmas make sense, since visually, the upper bound of Ric limit the velocity that the metric g changes. Therefore, we can derive a inclusion relationship between balls at different times.

4.1 A volume lower bound

Suppose $(M, g(t))$ is a complete Ricci flow for $t \in [0, +\infty)$ and $x_0 \in M$, with

$$\text{Ric}_{g(t)} \geq 0, \quad |K|_{g(t)} \leq c_0 t^{-1}.$$

For some $t_0 = t_0(n, c_0)$, consider the pullback $h(t) = \tau^* g(t)$ on a ball $\mathbb{B}_\pi$. The Ricci and sectional curvature conditions continue to hold, and

$$\text{Vol } B_{h(t)}(0, r) \geq v_0 r^n$$

for every $r \in (0, 1]$ and $t \in [0, c_0]$.

Indeed, by Lemma 4.2, there exists $r_0 > 0$ such that

$$B_{h(t_0)}(0, r_0) \subset B_{h(s)}(0, 1).$$

Consequently,

$$\begin{aligned} \text{Vol}_{h(s)} B_{h(s)}(0, 1) &\geq \text{Vol}_{h(t_0)} B_{h(s)}(0, 1) \\ &\geq \text{Vol}_{h(t_0)} B_{h(t_0)}(0, r_0) \\ &= \text{Vol}_{h(t_0)} \mathbb{B}_{r_0} \geq v_0. \end{aligned}$$

The desired estimate follows from Bishop–Gromov comparison. This relationship is essential because, in some cases, it implies a positive asymptotic volume ratio. We also have

$$B_{h(t)}(0, 1) \Subset \mathbb{B}_\pi.$$

4.2 Harnack conclusions

Theorem 4.3 (Harnack conclusions under mild curvature hypotheses). *Suppose $n \geq 3$, $(M^n, g(t))$ is a complete Ricci flow for $t \in [0, \infty)$, and $x_0 \in M$. Suppose that there exists $c_0 > 0$ such that*

- (A) $\text{Ric}_{g(t)} \geq 0$ for all $t \in [0, \infty)$;
- (B) $|K|_{g(t)} \leq c_0 t^{-1}$ for all $t \in (0, \infty)$.

Then the following two conclusions hold independently.

- (i) *If $\text{Ric}_{g(0)}(x_0) > 0$, then*

$$\liminf_{t \rightarrow \infty} t \text{Scal}_{g(t)}(x_0) > 0. \tag{4.1}$$

(ii) If there exists $\varepsilon \in \left(0, \frac{1}{n(n-1)}\right)$ such that

$$R_{g(t)} - \varepsilon \operatorname{Scal}_{g(t)} I \in C_{\text{PIC1}} \quad (4.2)$$

for all $t \geq 0$ and throughout M , then

$$\liminf_{t \rightarrow \infty} t \operatorname{Scal}_{g(t)}(x_0) = 0. \quad (4.3)$$

Proof. Define

$$\lambda := \liminf_{t \rightarrow \infty} t \operatorname{Scal}_{g(t)}(x_0).$$

By hypotheses (A) and (B), $\lambda \in [0, \infty)$. Choose $t_m \rightarrow \infty$ such that

$$t_m \operatorname{Scal}_{g(t_m)}(x_0) \rightarrow \lambda.$$

We might be tempted to apply the lifted-flow construction directly to $g(t)$. Instead, for each $m \in \mathbb{N}$, define a blown-down Ricci flow

$$g_m(t) := \frac{1}{R_m^2} g(R_m^2 t), \quad R_m := \left(\frac{t_m}{c_0}\right)^{1/2}.$$

This flow still satisfies (A) and (B), while

$$\begin{aligned} \operatorname{Scal}_{g_m(c_0)}(x_0) &= \operatorname{Scal}_{R_m^{-2}g(R_m^2 c_0)}(x_0) \\ &= R_m^2 \operatorname{Scal}_{g(R_m^2 c_0)}(x_0) \\ &= \frac{1}{c_0} t_m \operatorname{Scal}_{g(t_m)}(x_0) \longrightarrow \frac{\lambda}{c_0}. \end{aligned} \quad (4.4)$$

We apply the lifting construction to $g_m(t)$ instead. After discarding finitely many terms, we may assume that $R_m > 1$ for every $m \in \mathbb{N}$, so that $g_m(t)$ is genuinely a blow-down rather than a blow-up.

Applying the lifting argument to $g_m(t)$ gives a sequence of lifted flows $h_m(t)$ satisfying

- (a) $\operatorname{Ric}_{h_m(t)} \geq 0$ for all $t \geq 0$;
- (b) $|K|_{h_m(t)} \leq c_0 t^{-1}$ for all $t > 0$;
- (c) $B_{h_m(t)}(0, 1) \Subset \mathbb{B}_\pi$ for all $t \in [0, c_0]$;
- (d) $\operatorname{Vol} B_{h_m(t)}(0, r) \geq v_0 r^n$ for all $r \in (0, 1]$ and $t \in [0, c_0]$.

Passing to a subsequence and taking a limit gives a flow $h_\infty(t)$ such that

- (i) $\operatorname{Ric}_{h_\infty(t)} \geq 0$ for all $t \in (0, c_0]$;
- (ii) $|K|_{h_\infty(t)} \leq c_0 t^{-1}$ for all $t \in (0, c_0]$;
- (iii) $B_{h_\infty(t)}(0, s) \Subset \mathbb{B}_1$ for all $t \in (0, c_0]$ and $s \in (0, 1)$;
- (iv) $\operatorname{Vol} B_{h_\infty(t)}(0, r) \geq v_0 r^n$ for all $r \in (0, 1)$ and $t \in (0, c_0]$;
- (v) $\operatorname{Scal}_{h_\infty(c_0)}(0) = \lambda/c_0$.

Now blow up the flow by setting

$$H_m(t) := m^2 h_\infty(m^{-2}t).$$

The preceding properties become

- (I) $\text{Ric}_{H_m(t)} \geq 0$ for all $t \in (0, m^2 c_0]$;
- (II) $|K|_{H_m(t)} \leq c_0 t^{-1}$ for all $t \in (0, m^2 c_0]$;
- (III) $B_{H_m(t)}(0, s) \Subset \mathbb{B}_1$ for all $t \in (0, m^2 c_0]$ and $s \in (0, m)$;
- (IV) $\text{Vol } B_{H_m(t)}(0, r) \geq v_0 r^n$ for all $r \in (0, m)$ and $t \in (0, m^2 c_0]$.

Taking another limit gives a limit manifold $\mathcal{M}$ and a flow $H_\infty(t)$ satisfying

- (I) $\text{Ric}_{H_\infty(t)} \geq 0$ throughout $\mathcal{M}$ for all $t > 0$;
- (II) $|K|_{H_\infty(t)} \leq c_0 t^{-1}$ for all $t > 0$;
- (III) $\text{Vol } B_{H_\infty(t)}(0, r) \geq v_0 r^n$ for all $r > 0$ and $t > 0$, and therefore

$$\text{AVR}(H_\infty(t)) \geq \frac{v_0}{\omega_n} > 0$$

for all $t > 0$.

Proof of part (i).

Suppose, toward a contradiction, that the scalar-curvature limit is zero. Then

$$\text{Scal}_{h_\infty(c_0)}(0) = 0.$$

The evolution equation

$$\left(\frac{\partial}{\partial t} - \Delta_{h_\infty(t)} \right) \text{Scal}_{h_\infty(t)} = 2 |\text{Ric}_{h_\infty(t)}|^2$$

and the maximum principle imply that $\text{Ric} \equiv 0$. Hence the flows H_m and H_∞ are Ricci-flat and therefore static.

We now use an η -Ricci-ball argument. Without loss of generality, after rescaling the initial metric, assume that

$$\text{Ric}_{g_0} \geq \eta > 0$$

on $B_{g_0}(x_0, 1)$. This becomes

$$\text{Ric}_{g_m(0)} \geq \eta R_m^2$$

on $B_{g_m(0)}(x_0, R_m^{-1})$, and then

$$\text{Ric}_{h_m(0)} \geq \eta R_m^2$$

on $B_{h_m(0)}(0, R_m^{-1})$.

The Ricci lower bound forces slower volume growth than on Euclidean space. Consequently, there exists $\delta > 0$ such that

$$\text{Vol } B_{h_m(0)}(0, r) \leq (1 - 2\delta)\omega_n r^n, \quad r \in [R_m^{-1}, 1).$$

After flowing for a short time, this gives

$$\text{Vol } B_{h_m(t)}(0, r) \leq (1 - \delta)\omega_n r^n, \quad r \in [R_m^{-1}, 1), \quad t \in [0, \rho r^2].$$

Passing to the limit yields

$$\text{Vol } B_{h_\infty(t)}(0, r) \leq (1 - \delta)\omega_n r^n, \quad r \in (0, 1), \quad t \in (0, \rho r^2]. \quad (*)$$

Here we cannot take $t = 0$, because $t = 0$ is the singular time.

The aim is to let $r \rightarrow 0$ and contradict the asymptotic behavior of volume at the origin. The restriction $t \leq \rho r^2$ prevents us from fixing a uniform positive time during this limiting process. The staticity of $H_\infty(t)$ resolves this problem.

Indeed, $(*)$ rescales to

$$\text{Vol } B_{H_m(t)}(0, r) \leq (1 - \delta)\omega_n r^n, \quad r \in (0, m), \quad t \in (0, \rho r^2].$$

Hence

$$\text{Vol } B_{H_\infty(t)}(0, r) \leq (1 - \delta)\omega_n r^n$$

for $r > 0$ and $t \in (0, \rho r^2]$. Since $H_\infty(t) = H_\infty(1) =: H$, we obtain

$$\text{Vol } B_H(0, r) \leq (1 - \delta)\omega_n r^n$$

for every $r > 0$. Letting $r \rightarrow 0$ gives a contradiction.

Proof of part (ii).

Suppose the limit is nonzero. The pinching condition holds, and we have already shown that $\text{AVR} > 0$. The result of Deruelle, Schulze, and Simon therefore implies that the limit manifold is flat, contradicting the nonzero limit. $\square$

Finally, [theorem 1.1](#) follows readily from the Ricci-flow existence theorem. If the initial manifold is not flat, we can find a point at which $\text{Ric} > 0$. At that point, parts (i) and (ii) of [theorem 4.3](#) imply respectively that

$$\liminf_{t \rightarrow \infty} t \text{Scal}_{g(t)}(x_0) > 0 \quad \text{and} \quad \liminf_{t \rightarrow \infty} t \text{Scal}_{g(t)}(x_0) = 0,$$

a contradiction.

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